

SARMAE: Masked Autoencoder for SAR Representation Learning

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Abstract

Synthetic Aperture Radar (SAR) imagery plays a critical role in all-weather, day-and-night remote sensing applications. However, existing SAR-oriented deep learning is constrained by data scarcity, while the physically grounded speckle noise in SAR imagery further hampers fine-grained semantic representation learning. To address these challenges, we propose SARMAE, a Noise-Aware Masked Autoencoder for self-supervised SAR representation learning. Specifically, we construct SAR-1M, the first million-scale SAR dataset, with additional paired optical images, to enable large-scale pre-training. Building upon this, we design Speckle-Aware Representation Enhancement (SARE), which injects SAR-specific speckle noise into Masked Autoencoder to facilitate noise-aware and robust representation learning. Furthermore, we introduce Semantic Anchor Representation Constraint (SARC), which leverages paired optical priors to align SAR features and ensure semantic consistency. Extensive experiments across multiple SAR datasets demonstrate that SARMAE achieves state-of-the-art performance on classification, detection, and segmentation tasks. Code, data, and models available at [SARMAE](#).

1. Introduction

Synthetic Aperture Radar (SAR) imagery, characterized by its all-weather, day-and-night imaging capability, serves as a critical source of remote sensing information [2, 4, 20, 28, 60, 98]. Owing to this robustness against challenging atmospheric and illumination variations, SAR has been widely applied in various domains such as ocean

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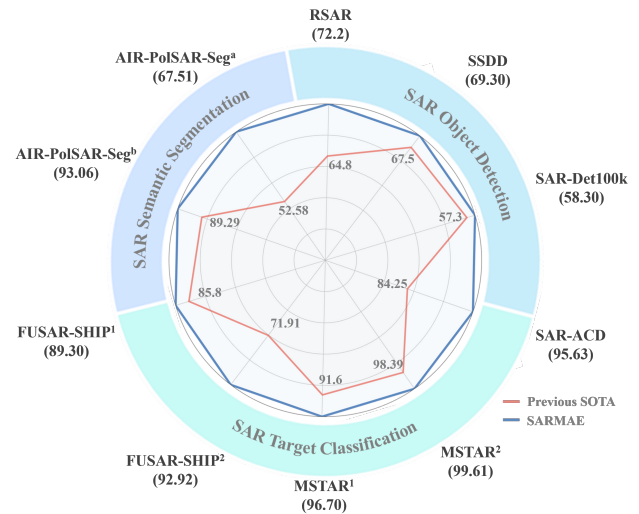


Figure 1. SARMAE outperforms SOTA methods on multiple datasets. ¹: 40-SHOT; ²: 30% labeled. ^a: Multi-classes; ^b: Water.

monitoring, disaster assessment, and urban scene analysis [46, 48, 49, 51, 59, 80, 83].

The emergence of deep learning has greatly advanced SAR image interpretation [96]. Early approaches were typically designed for specific tasks [31], exhibiting limited generalization across diverse downstream applications. To overcome this limitation, subsequent representation learning methods adopted the pretraining–finetuning paradigm [15, 33, 34, 43, 81] to achieve unified and generalized interpretation. In this context, self-supervised pretraining methods [1, 7, 25] have shown great potential for learning transferable representations from unlabeled data, thereby inspiring the development of analogous frameworks tailored for SAR imagery [16, 35].

Despite these advances, existing SAR representation learning methods still face several limitations. First, due to the high cost of SAR data acquisition, current studies often lack sufficiently large and diverse pretraining datasets, limiting the ability of models to learn general-purpose SAR representations [11, 29, 57, 62–64]. Second, the unique imaging mechanism of SAR introduces severe speckle noise, leading to low semantic content and weak structural cues [10, 28, 50, 68–70]. However, prevailing pre-training methods simply adopt strategies borrowed from optical imagery [15], overlooking the physical nature of SAR noise. Third, pretraining paradigms that rely exclusively on SAR data are inherently limited by its semantic scope [8, 37, 52]. This overlooks the valuable guidance that could be drawn from complementary, information-rich modalities such as optical imagery [19]. Consequently, the learned representations often lack the semantic richness and generalizability required for diverse downstream tasks.

To address these challenges, we propose SARMAE, a Masked Autoencoder (MAE) [25]-based framework for SAR representation learning. Specifically, we construct SAR-1M, the first million-scale SAR dataset that covers diverse scenes, targets, and sensor characteristics, providing over 1.3 million SAR images for large-scale pretraining. We then introduce Speckle-Aware Representation Enhancement (SARE), which explicitly incorporates SAR imaging noise during pretraining. By reconstructing the original image from perturbed inputs injected with SAR-specific speckle noise, SARE encourages the network to perceive and capture the intrinsic speckle characteristics of SAR imagery, thereby learning noise-resistant representations and improving robustness. Finally, we design Semantic Anchor Representation Constraint (SARC). Considering the more discernible semantic structure of optical imagery, SARC aligns SAR features with geographically paired optical priors to guide the encoder learning. This alignment enforces semantic consistency, accelerates model convergence, and enhances the generalization of pretrained representations. Extensive experiments on various downstream tasks demonstrate that SARMAE achieves state-of-the-art performance across multiple SAR datasets, as illustrated in Fig. 1. Our main contributions are summarized as follows:

- We propose SAR-1M, the first million-scale SAR dataset covering diverse resolutions, scenes, scales, targets, and task scenarios, along with additional paired optical images, enabling large-scale pretraining.
- We introduce Speckle-Aware Representation Enhancement to enforce robust feature learning by reconstructing inputs corrupted with physically grounded speckle noise, thereby mitigating the adverse impact of noise on SAR feature extraction.
- We design Semantic Anchor Representation Constraint to align SAR features with paired optical priors to regular-

ize encoder learning, ensuring semantic consistency and enhancing representation generalization.

2. Related work

2.1. SAR Datasets for Representation Learning

Many SAR datasets are characterized by extreme specialization, either focusing on single-category targets such as MSTAR [12] or SSDD [88], or being curated for a single downstream task, as exemplified by the detection dataset SAR-Det100k [38]. Beyond this narrow scope lies a more fundamental limitation: the lack of scale across the community’s available resources. Consequently, even state-of-the-art pretrained models such as SARATR-X [35] and SUMMIT [16] are trained on relatively small datasets containing only 180k and 560k images, respectively. This combination of limited semantic diversity, task specificity, and insufficient data volume imposes a clear ceiling on model performance and hinders the development of generalizable pretraining paradigms for SAR representation learning. To this end, we introduce SAR-1M, the first million-scale SAR dataset spanning diverse resolutions, imaging scales, object categories, and task scenarios, providing the scale and semantic richness required for large-scale pretraining.

2.2. SAR Representation Learning

Recently, large-scale self-supervised pretraining paradigms have been introduced to remote sensing [21, 66, 67, 71]. SeCo [47] adopts a MoCo-style [24] framework for scene classification, while RVSA [62] leverages MAE [25] pretraining to obtain initial weights and further introduces rotated varied-size window attention to enhance representation learning. SatMAE [9] extends MAE to multispectral and multitemporal imagery, and ScaleMAE [53] introduces scale-aware masking to improve generalization across spatial resolutions. Other approaches aim to learn task-agnostic representations by leveraging massive unlabeled datasets across diverse modalities [18, 26, 58, 65, 79]. However, these efforts remain predominantly focused on optical imagery, overlooking the unique characteristics of SAR data—particularly its inherent speckle noise and distinct imaging physics.

Motivated by the success of unsupervised pretraining paradigms in optical imagery, several studies have explored self-supervised SAR representation learning. BIDFC [85] adopts a contrastive learning framework with weak supervision, while SAR-JEPA [32] leverages masked autoencoding and local reconstruction to extract spatial features. MSFA [38] further introduces a multi-stage filter-based strategy. Although these approaches provide valuable insights, they remain constrained to single-task settings. To address these limitations, recent works have proposed unified pretraining frameworks tailored for SAR imagery.

Model	Pretrain Dataset	Modality	Task Types	Size	Targets	Resolution	Band
MSFA [38]	DOTA [76]	OPT	Det.	-	-	-	-
SAR-JEPA [32]	SAR-JEPA	SAR	Cls.	90k	21	0.3m-25m	C/X
SARATR-X [35]	SAR-VSA	SAR	Cls.&Det.	180k	25	0.1m-25m	C/X/Ku/Ka
SUMMIT [16]	MuSID	SAR	Cls.&Det.	560k	41	0.1m-25m	C/X
SARMAE	SAR-1M	SAR&OPT	Cls.&Det.&Seg.	1.3M SAR/2.3M Total*	57	0.1m-60m	C/X/Ku/Ka

Table 1. Pretraining datasets for SAR Representation Learning. *Our dataset comprises 1.3M SAR and 1M paired OPT images.

SARATR-X [35] employs a HiViT [89] with a two-stage self-supervised learning framework for unified SAR target detection and classification, while SUMMIT [16] adopts a masked image modeling framework with multiple auxiliary self-supervised tasks to adapt to diverse downstream scenarios. These two models demonstrate the feasibility of learning general-purpose SAR representations through large-scale pretraining. However, both methods overlook the intrinsic physical priors of SAR imaging and rely solely on low-quality, noise-contaminated data for pretraining, which limits the semantic richness of their learned representations. In this paper, we propose SARMAE, a noise-aware framework that explicitly models realistic speckle characteristics during pretraining and incorporates high-fidelity paired optical imagery as semantic guidance, thereby enhancing the model’s capability to learn robust and semantically consistent representations from complex SAR data.

3. Method

In this section, we first introduce the constructed SAR-1M dataset. Based on this foundation, we then present SAR-MAE, which incorporates two key innovations: Speckle-Aware Representation Enhancement (SARE) and Semantic Anchor Representation Constraint (SARC).

3.1. SAR-1M Dataset

Existing public datasets for SAR image analysis are hampered by limitations in scale and task diversity. To overcome these obstacles, we construct a million-scale, multi-source dataset, termed SAR-1M.

SAR-1M is a curated aggregation of 18 publicly available datasets, as illustrated in Fig. 2. During data collection, we adhered to two core principles: semantic richness and sensor diversity. For semantic richness, we selected datasets covering representative object categories such as ship, aircraft, vehicle, and bridge, as well as land-cover types including bareland, rangeland, agricultural land, and mountains. In addition, SAR-1M incorporates event-related scenes (*e.g.*, earthquake), further enriching the contextual variety. Notably, many of these datasets were originally designed for detection or segmentation tasks, meaning that their samples typically have larger spatial coverage and richer fine-grained semantics. As a result, SAR-1M encom-

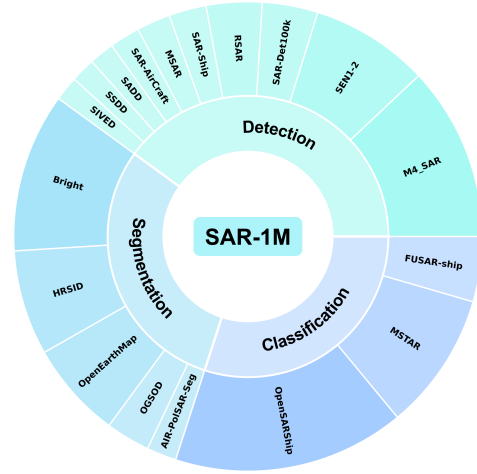


Figure 2. The organization of data sources in SAR-1M.

passes over 50 distinct categories, offering extensive semantic diversity and enhancing the representation capability of SAR-oriented models.

Beyond semantic diversity, SAR-1M integrates imagery from multiple sensors: Sentinel-1, Gaofen-3, RadarSat-2, Terra A/S, and TerraSAR-X, covering multiple frequency bands (C, X, Ku, Ka) and polarization modes (HH, HV, VV, VH). These diverse configurations yield data with varying ground resolutions and scattering characteristics, enriching the representation of different imaging conditions. Moreover, as several source datasets were collected on a global scale (*e.g.*, SEN1-2 [55]), SAR-1M exhibits remarkable geographical diversity, which further enhances the generalization and adaptability of pretrained models.

Unlike existing SAR-only datasets, SAR-1M also incorporates geographically aligned SAR–optical pairs, resulting in approximately 2.3 million samples. Even when restricted to the SAR modality, it still comprises over 1.3 million images, providing ample scale for robust pretraining.

It is worth noting that the number of channels in the originally collected SAR images varies across datasets. To ensure consistency for unified pretraining, each polarization mode was expanded to a three-channel format when the original image contained a number of channels not equal to three. A detailed comparison between SAR-1M and existing SAR pretraining datasets is presented in Table 1.

3.2. SARMAE

3.2.1. Overall framework

Fig. 3 illustrates the overall framework of SARMAE for pretraining, which consists of a SAR branch and an optical branch. The SAR branch follows the architecture of MAE [25], where the encoder adopts a ViT backbone and the decoder is composed of multiple Transformer layers same as the original MAE. The optical branch employs a frozen DINOv3 [56] network that shares the same ViT-based encoder architecture as the SAR branch, facilitating subsequent feature alignment between the two modalities.

During pretraining, a SAR image is fed into the SAR branch, following the same procedure as MAE. To cope with the speckle noise inherent in SAR imagery, we explicitly introduce it into the SAR branch and enforce the network to learn effective denoising representations. This mechanism, termed Speckle-Aware Representation Enhancement (SARE), improves the model's ability to capture robust and discriminative SAR features. Furthermore, considering that some SAR images are paired with optical counterparts, we propose a Semantic Anchor Representation Constraint (SARC) to fully exploit such multimodal correspondence. In this design, the optical image is passed through the frozen DINOv3 branch to obtain discriminative semantic features, which serve as semantic anchors to guide the SAR encoder's representation learning. This constraint not only facilitates model convergence but also enhances generalization capability.

It is worth noting that if a SAR image does not have a paired optical counterpart, it is only processed by the SAR branch. The following sections provide detailed explanations of SARE and SARC.

3.2.2. Speckle Aware Representation Enhancement

Unlike additive Gaussian noise, speckle is a multiplicative phenomenon arising from the coherent summation of backscattered signals within a resolution cell. For a multi-look SAR intensity image, the observed intensity Z is the result of incoherently averaging L independent single-look measurements $\{I_i\}_{i=1}^L$, where each I_i follows an exponential distribution with mean $\bar{I}$ equal to the true backscattering intensity. This process is formally described as:

$$Z = \frac{1}{L} \sum_{i=1}^L I_i, \quad \text{where } I_i \sim \text{Exp}(\bar{I}), \quad E[I_i] = \bar{I}. \quad (1)$$

Consequently, the multi-look intensity Z follows a Gamma distribution, with its probability density function (PDF) given by:

$$p_Z(z|\bar{I}, L) = \frac{L^L}{\Gamma(L)\bar{I}^L} z^{L-1} \exp\left(-\frac{Lz}{\bar{I}}\right), \quad z \geq 0, \quad (2)$$

where $\Gamma(\cdot)$ is the Gamma function, L is the number of looks, and the mean of Z is $E[Z] = \bar{I}$. Equation 2 provides

a physically grounded model for speckle, explicitly showing that the observed pixel intensity z is a random variable drawn from a distribution whose shape is determined by the true scene signal $\bar{I}$. This statistical entanglement is precisely what confuses standard pre-training paradigms and motivates our physics-informed design.

Grounded in this statistical model, we design SARE. Instead of merely reconstruct the original (and already noisy) masked patches, we reformulate the self-supervised task so that the model is trained to denoise a synthetically corrupted version of the input back to the original SAR patch. Specifically, for any given input SAR patch x , it is treated as the best available estimate of the true signal (*i.e.*, $x \approx \bar{I}$). A more heavily corrupted observation x' is then synthesized by sampling from the Gamma distribution with a reduced synthetic look number L_{syn} :

$$x'(i, j) \sim \text{Gamma}\left(L_{\text{syn}}, \frac{x(i, j)}{L_{\text{syn}}}\right). \quad (3)$$

This construction preserves the pixel-wise mean $E[x'(i, j)] = x(i, j)$ while increasing the variance through the lower look number. Here, L_{syn} is chosen to be significantly smaller than the effective number of looks in the original patch, ensuring that the synthesized patch x' exhibits a higher noise level.

The pre-training task is then formulated as a denoising reconstruction objective. Given the synthetically corrupted patch x' , we apply random masking with a high masking ratio (75% following MAE) to obtain a masked version $\tilde{x}'$, retaining only a subset of visible patches indexed by $\mathcal{V}$. The visible patches are fed to the encoder E_{SAR} to produce latent representations, which are then processed by the decoder D along with learnable mask tokens to reconstruct the original patch x . The reconstruction loss of SARE is defined as the mean squared error (MSE) between the predicted patches and the corresponding original patches in the masked regions $\mathcal{M}$:

$$\mathcal{L}_{\text{SARE}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \|D(E_{\text{SAR}}(\tilde{x}'))_p - x_p\|_2^2, \quad (4)$$

where p indexes the masked patch positions, $|\mathcal{M}|$ denotes the number of masked patches, $D(E_{\text{SAR}}(\tilde{x}'))_p$ represents the reconstructed patch at position p , and x_p is the corresponding ground-truth patch from the original input. This formulation trains the model to map noisy inputs to clean and complete representations. By doing so, the encoder learns features that are robust to the statistical fluctuations inherent to SAR imaging and are largely disentangled from speckle noise.

Notably, in our implementation, in addition to the Gamma model, we also incorporate Rayleigh, Gaussian, and Uniform noise, to generate more diverse corrupted samples and further improve model robustness.

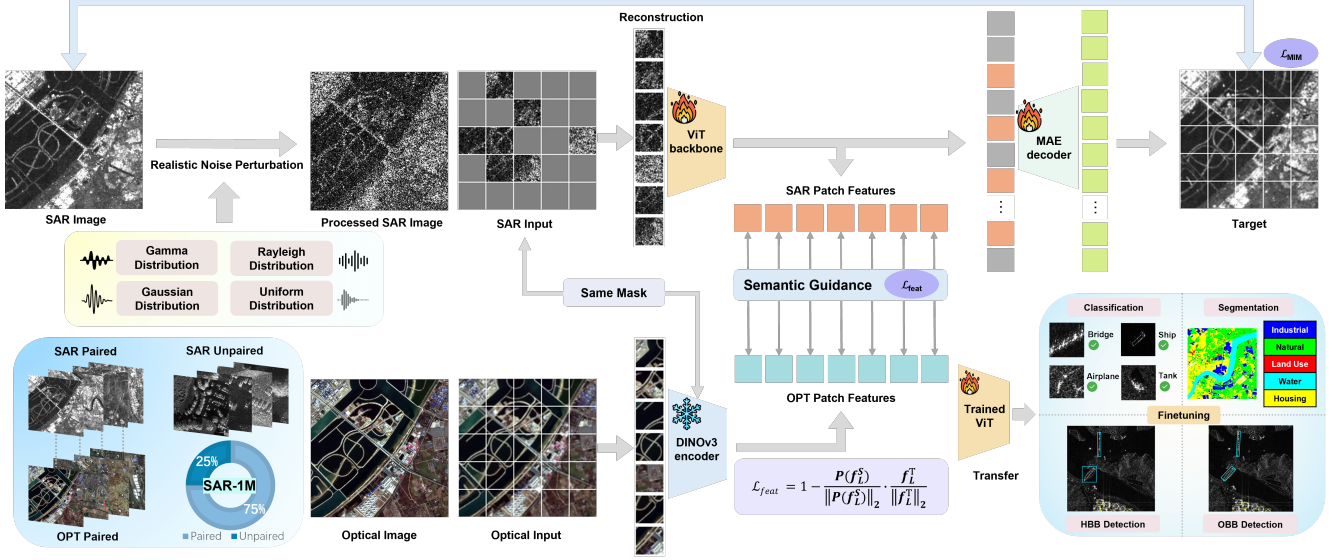


Figure 3. Overview of the SARMAE pretraining framework. The framework consists of two branches: (i) a SAR branch following the MAE architecture with Speckle-Aware Representation Enhancement (SARE) to handle inherent speckle noise, and (ii) an optical branch using a frozen DINOv3 encoder. For paired SAR-optical data, Semantic Anchor Representation Constraint (SARC) aligns SAR features with semantic-rich optical representations. Unpaired SAR images are processed solely through the SAR branch.

3.2.3. Semantic Anchor Representation Constraint

While SARE improves robustness to speckle noise, it does not provide the high-level semantic cues that are readily available in paired optical imagery. To this end, we introduce the SARC.

For a given co-registered image pair (I_{SAR}, I_{OPT}) , the SAR image is masked using the MAE strategy and passed through E_{SAR} to obtain embeddings for the visible patches,

$$f_{SAR} = \{f_{SAR}^i\}_{i \in \mathcal{V}}, \quad (5)$$

where $\mathcal{V}$ denotes the set of visible patch indices. Concurrently, the unmasked optical image is processed by the frozen encoder E_{OPT} to produce a full sequence of anchor embeddings,

$$f_{OPT} = \{f_{OPT}^i\}_{i=1}^N. \quad (6)$$

Then, we enforce that the representation of a SAR patch should be semantically aligned with its spatially corresponding optical patch. This is achieved through a patch-wise cosine distance loss. For each visible SAR patch embedding f_{SAR}^i at spatial position $i \in \mathcal{V}$, the optical patch embedding f_{OPT}^i at the same location serves as the semantic target. Accordingly, the SARC loss is defined as the mean cosine distance over all visible patches:

$$\mathcal{L}_{SARC} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \left(1 - \frac{f_{SAR}^i \cdot f_{OPT}^i}{\|f_{SAR}^i\|_2 \|f_{OPT}^i\|_2} \right). \quad (7)$$

This objective directly minimizes the cosine distance between the SAR and optical feature vectors, enforcing strong

directional alignment in the embedding space. As a result, the SAR encoder E_{SAR} learns not only SAR-specific structural cues, but also embeds them in a feature space shaped by the discernible semantic structure provided by the frozen optical teacher.

Overall Pre-training Objective. The overall pre-training loss consists of two components: the SARE loss and the SARC loss:

$$\mathcal{L}_{pretrain} = \mathcal{L}_{SARE} + \lambda \mathcal{L}_{SARC}, \quad (8)$$

where λ balances the modality-specific reconstruction objective and the cross-modal semantic alignment. In our implementation, we set $\lambda = 0.1$. Through this dual-objective optimization, the SAR encoder learns representations that are simultaneously robust to speckle noise and enriched with meaningful semantics, enabling superior performance across downstream perception tasks.

4. Experiments

In this section, we conduct comprehensive experiments to evaluate the proposed method across multiple downstream tasks, including SAR target classification, object detection, and semantic segmentation. We also assess the effectiveness of the SAR-1M dataset and the contributions of the proposed SARE and SARC modules through ablation studies. We initialize the SAR-branch encoder with ViT weights pretrained on ImageNet [11] using MAE [25]. The model is trained for 300 epochs using AdamW (learning rate =

Method	FUSAR-SHIP		MSTAR		SAR-ACD
	40-shot	30%	40-shot	30%	30%
ResNet-50 [23]	-	58.41	-	89.94	59.70
Swin Transformer [41]	-	60.79	-	82.97	67.50
Beit [1]	59.70	71.13	40.70	69.75	79.77
LoMaR [36]	82.70	-	77.00	-	-
SAR-JEPA [32]	85.80	-	91.60	-	-
SUMMIT [16]	-	71.91	-	98.39	84.25
Copernicus FM [73]	87.61	-	-	-	92.63
CROMA [18]	83.71	-	-	-	88.99
SARMAE(ViT-B)	89.30	92.92	96.70	99.61	95.06
SARMAE(ViT-L)	90.86	92.80	97.24	98.92	95.63

Table 2. Performance comparison (Top1 Accuracy, %) of different methods on the target classification task.

Method	SARDet-100k	SSDD	Method	RSAR
ImageNet [11]	52.30	66.40	RoI-Transformer [13]	35.02
Deformable DETR [97]	50.00	52.60	Def. DETR [97]	46.62
Swin Transformer [41]	53.80	40.70	RetinaNet [39]	57.67
ConvNeXt [42]	55.10	-	ARS-DETR [84]	61.14
CATNet [14]	-	64.66	R3Det [82]	63.94
MSFA [38]	56.40	-	ReDet [22]	64.71
SARATR-X [35]	57.30	67.50	O-RCNN [87]	64.82
SARMAE(ViT-B)	57.90	68.10	SARMAE(ViT-B)	66.80
SARMAE(ViT-L)	63.10	69.30	SARMAE(ViT-L)	72.20

Table 3. Performance comparison (mAP, %) of different methods on horizontal and oriented object detection tasks.

1×10^{-3} , weight decay = 0.05) with a batch size of 1024 and a cosine learning-rate schedule. All the experiments are performed on NVIDIA A800 GPU. All fine-tuning details are provided in the supplementary material.

4.1. Main Results

We compare SARMAE with existing state-of-the-art methods. In addition to general-purpose visual representation models such as ImageNet-pretrained MAE [25], BEiT [1], and Swin Transformer [41], we also include the most recent SAR-specific approaches, including SAR-JEPA [32], SARATR-X [35], MSFA [38], and SUMMIT [16].

The scene classification and object detection comparison results are obtained from SAR-JEPA [32], SARATR-X [35], and SUMMIT [16], while the semantic segmentation results are directly sourced from AIR-PoSAR-Seg [95]. The downstream performance of ViT-B and ViT-L after pretraining is presented in the corresponding tables.

Target Classification. We first evaluate SARMAE on the target classification task, which does not require an additional decoder and therefore provides a direct assessment of the model’s representation capability. In implementation, the feature map from the last ViT layer is processed by global average pooling and then passed through a linear classifier to produce logits corresponding to the number of classes. We conduct experiments on three widely used datasets: FUSAR-SHIP [27], MSTAR [12],

and SAR-ACD [91]. We compare the performance of SARMAE with Copernicus FM [73] and CROMA [18], which adopts DINO distillation and contrastive learning respectively. Table 2 shows superior performance of SARMAE on FUSAR and SAR-ACD. Then, following the protocols in [16, 32], we adopt the 40-shot and 30% label settings for the FUSAR-SHIP and MSTAR datasets. As shown in Table 2, SARMAE consistently achieves state-of-the-art performance across all evaluation settings. For example, under the 30% label setting on FUSAR-SHIP, our ViT-B model reaches 92.92% accuracy, outperforming the previous best by a large margin of 21.01%. These results clearly demonstrate the effectiveness of the our approach.

Horizontal & Oriented Object Detection. We then evaluate the transferability of the learned representations on the object detection task. For horizontal object detection, we use the classical SSDD [88] dataset and the more challenging SARDet-100k [38]. For oriented bounding box detection, we adopt the recently proposed large-scale RSAR dataset [90]. Following the fine-tuning protocols of [35], we integrate our pretrained backbones into Faster R-CNN [54] and Oriented R-CNN [87] for horizontal and oriented detection, respectively. The results are detailed in Table 3.

In the horizontal object detection scenario represented by SARDet-100k, our model with a standard ViT-B backbone achieves 57.9 mAP, slightly outperforming SARATR-X (built on HiViT-B) which reports 57.3 mAP. This result is particularly noteworthy because HiViT [89], with its hierarchical and pyramidal design, is generally regarded as an improved variant of ViT and typically delivers stronger performance. Nevertheless, our ViT-based model still attains the best results, underscoring that the performance gain primarily arises from the superior representations learned by SARMAE, rather than architectural advantages.

In the oriented object detection task, SARMAE also secures top performance with 66.8% mAP, surpassing existing methods by a large margin. Notably, our model exhibits excellent scalability: upgrading the backbone from ViT-B to ViT-L further boosts performance by +5.4 mAP, thereby widening the gap over previous approaches.

Semantic Segmentation. Beyond image-level and object-level evaluation, we further assess the pretrained model on pixel-level tasks through semantic segmentation. For this purpose, we adopt the well-known AIR-PoSAR-Seg dataset [95], which includes two settings: multiclass segmentation and single-class water extraction. Our implementation follows the protocol in [95], using UperNet [78] as the segmentation framework. As shown in Table 4, SARMAE achieves clear and consistent advantages over existing methods. By incorporating optical cues through SARC during pretraining, SARMAE captures clearer semantic structures, such as boundaries and texture structures, yielding superior segmentation results, particularly for the more chal-

lenging multiclass setting.

Method	Multiple classes							Water
	Industrial Area	Natural Area	Land Use	Water	Housing	Other	mIoU	IoU
FCN [44]	37.78	71.58	1.24	72.76	67.69	39.05	48.35	85.95
ANN [99]	41.23	72.92	0.97	75.95	68.40	56.01	52.58	87.32
PSPNet [92]	33.99	72.31	0.93	76.51	68.07	57.07	51.48	87.13
DeepLab V3+ [6]	40.62	70.67	0.55	72.93	69.96	34.53	48.21	87.53
PSANet [93]	40.70	69.46	1.33	69.46	68.75	32.68	47.14	86.18
DANet [17]	39.56	72.00	1.00	74.95	67.79	56.28	39.56	89.29
SARMAE(ViT-B)	65.87	75.65	29.20	84.01	73.23	71.21	66.53	92.31
SARMAE(ViT-L)	65.84	78.04	29.47	87.12	75.22	69.34	67.51	93.06

Table 4. Performance comparison of semantic segmentation methods on multiple classes and water classes.

4.2. Ablation Study

In this section, we investigate the effectiveness of the SAR-1M dataset and conduct ablation studies on SARMAE. We use a ViT-B model pretrained on ImageNet [11] with MAE [25] as our baseline to clearly highlight the contribution of each component in our method. All our ablation models training on SAR-1M use the same settings, i.e., 300 epochs, ensuring a controlled and fair comparison. For efficient verification, all ablation comparisons are conducted using the ViT-B backbone, and the effectiveness of each variant is demonstrated through fine-tuning on different downstream tasks.

Efficacy of SAR-1M. As shown in Table 5, we evaluate the impact of in-domain pretraining by applying the standard MAE framework across different datasets and fine-tuning on various SAR downstream tasks. It can be seen that, models pretrained from scratch on SAR-1M consistently outperform those pretrained on ImageNet or MillionAID. For example, on the FUSAR-SHIP classification benchmark, in-domain pretraining on SAR-1M improves accuracy from 75.40% (ImageNet) to 82.22%, yielding a substantial +6.82% absolute gain. Similarly, we observe improvements of +0.2 mAP on detection and +4.08 mIoU on segmentation. These results demonstrate that the unique characteristics of SAR imagery introduce a significant distribution gap from both natural and optical remote-sensing images, making it difficult for models pretrained on those modalities to transfer effectively. This underscores the necessity of constructing a large-scale SAR-specific pretraining dataset. Therefore, SAR-1M is not merely a large collection of images, but a high-quality, domain-aligned resource that facilitates the development of powerful representation models tailored for SAR imagery.

Ablation on SARE. We further validate the effectiveness of the SARE module by integrating it into the SAR-1M-pretrained MAE. As shown in Table 5, injecting physically grounded speckle noise yields consistent and substantial gains across all downstream tasks: classification accuracy on FUSAR-SHIP improves by +4.58%, and segmentation mIoU on AIR-PolSARSeg increases by +0.79%. These results support our central hypothesis that compelling the

Model	Pretrain Dataset	SARE	SARC	FUSAR	SSDD	AIR-PolSAR-Seg
MAE (Baseline)	ImageNet-1K [11]			75.40	64.00	60.28
MAE	MillionAID [45]			80.16	63.60	62.29
DINOv3	LVD-1689M [56]			74.25	61.60	58.42
MAE*	SAR-1M (only SAR)			82.22	64.20	64.36
MAE* (Add noise)	SAR-1M (only SAR)	✓		86.80	64.40	65.15
SARMAE*	SAR-1M (SAR/OPT)	✓	✓	89.30	68.10	66.53

Table 5. Ablation of SARE and SARC on multiple SAR datasets. *: Initialized with ImageNet pretrained MAE.

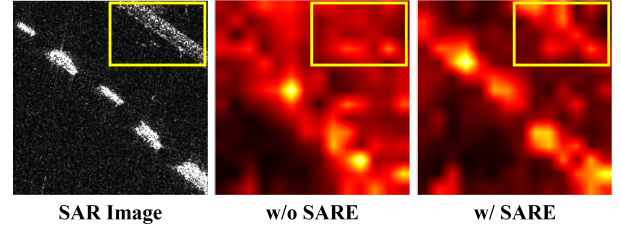


Figure 4. SARE significantly enhances the model’s semantic perception. Attention maps are computed by measuring the attention between the MAE encoder’s final-layer class token and image patch tokens.

model to reconstruct denoised targets enables it to better understand the noise characteristics of SAR imagery, thereby achieving noise-aware representation learning.

To provide more intuitive insights, we qualitatively examine how the model perceives scene content, as shown in Figure 4. With SARE, scene-relevant objects are more distinctly highlighted and exhibit clearer contours, indicating that the model can overcome noise interference and accurately capture semantic structures. Notably, the model is able to attend to subtle yet semantically relevant objects (see the yellow box), further indicating its enhanced ability to capture meaningful structures in noisy SAR imagery.

Ablation on SARC. Building upon MAE (add noise), we further evaluate the effectiveness of the SARC module by incorporating paired optical images during pretraining. As shown in Table 5, SARC yields consistent and significant improvements across all three downstream tasks. The effect is particularly notable on the SSDD detection benchmark, where performance increases by +3.7% (from 64.4% to 68.1%). In our experiments on this dataset, we observe that models often suffer from overfitting manifested as excessive false alarms, likely because the network misinterprets speckle noise as potential targets, indicating insufficient ability to capture true object semantics. By leveraging paired optical images, SARC provides clearer and more discriminative structural cues, enabling the model to better separate meaningful objects from noise. As a result, the detection accuracy improves substantially. Interestingly, although DINOv3 [56] has achieved remarkable performance on natural images, simply finetuning it on SAR images yields suboptimal results. This observation indicates that the effectiveness of SARC stems from explicit SAR-optical

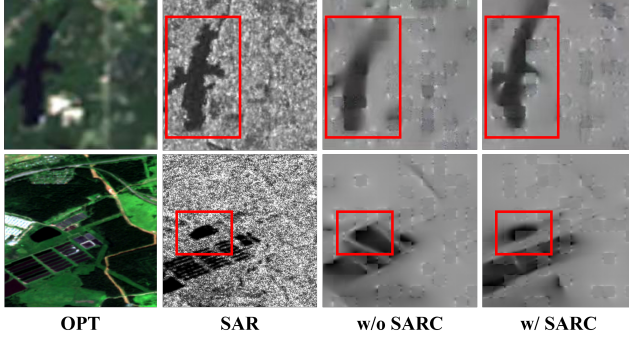


Figure 5. SARC leverages paired optical priors to recover the structural details of the original SAR image, which the model fails to reconstruct without this module.

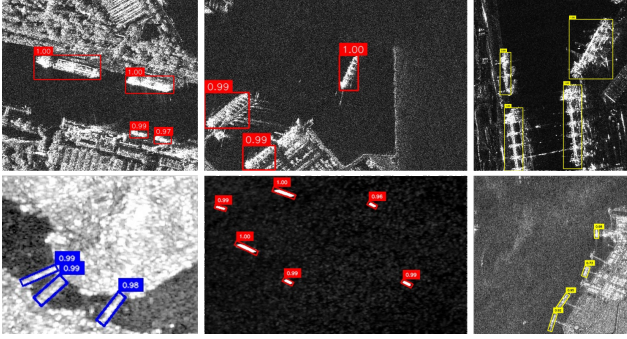


Figure 6. Visualization of Horizontal Object Detection Tasks on SSDD [88] and SAR-Det100k [38] in line 1. Oriented Object Detection Tasks on RSAR [90].

alignment, rather than from the strength of the optical backbone itself (i.e., DINOv3).

To further illustrate this mechanism, we visualize reconstructions from different pretrained MAE variants, as shown in Figure 5. With SARC, the model can recover the local scene structures present in the original SAR image (see the red box), whereas training on noisy SAR data alone results in blurred or incomplete reconstructions. This demonstrates that SARC injects clear optical semantic priors, enabling the model to discern true scene content and substantially enhancing the noise-aware masked image modeling process.

Based on the above ablation results, we find that SARE encourages the model to identify noise components irrelevant to scene understanding, enabling it to learn the correct information. In contrast, SARC introduces paired optical priors that guide the model toward clearer semantic structures. Together, these two components operate from complementary perspectives and jointly enhance the model’s representation and generalization ability on SAR imagery.

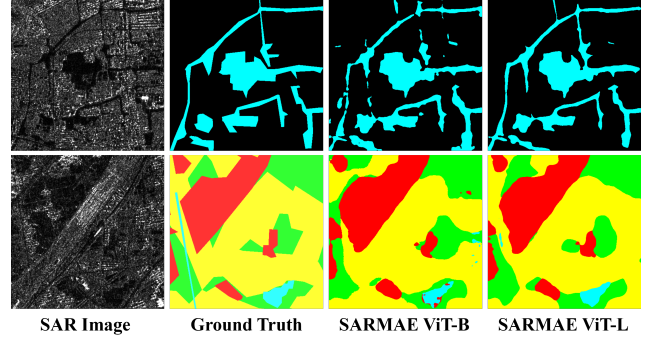


Figure 7. Visualization of segmentation tasks on AIR-PolSAR-Seg [95].

4.3. Visualization

Figure 6–7 present the qualitative results of our model on detection and segmentation tasks, respectively. As shown, our model can accurately detect targets of varying locations, shapes, and sizes in SAR imagery. In particular, for the challenging oriented detection scenario, our approach successfully captures the ship docked at the port with high confidence (see the yellow box in the second row of Figure 6), even though its appearance is difficult to distinguish from the surrounding environment. For single-class semantic segmentation, our model achieves robust performance: despite strong noise interference, it is still able to correctly identify water bodies with clear and coherent boundaries. In multi-class segmentation tasks, our method also produces segmentation maps of high visual quality. These results intuitively demonstrate the effectiveness of our approach.

5. Conclusion

In this paper, we present SARMAE, a self-supervised pre-training framework tailored for SAR representation learning. We introduce SAR-1M, the first million-scale SAR dataset with paired optical imagery, enabling large-scale pretraining directly aligned with SAR-specific characteristics. Building upon this, we propose two core innovations: Speckle-Aware Representation Enhancement (SARE), which explicitly models the statistical behavior of SAR speckle to learn noise-aware representations, and Semantic Anchor Representation Constraint (SARC), which leverages paired optical priors to guide SAR feature learning and improve semantic fidelity. Extensive experiments across classification, detection, and segmentation demonstrate that SARMAE achieves SOTA performance, validating both the quality of SAR-1M and the effectiveness of our pre-training framework. We believe that SAR-1M and SARMAE will constitute a strong foundation for future research in SAR representation learning and foster the development of SAR-oriented foundation models.

Acknowledgement

This work was supported in part by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (JYB2025XDXM101), the National Natural Science Foundation of China (624B2109, 623B2079, 62225113), the Zhongguancun Academy Project (20240308), and the Key Technology Research Project of China National Petroleum Corporation (2025ZG82).

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